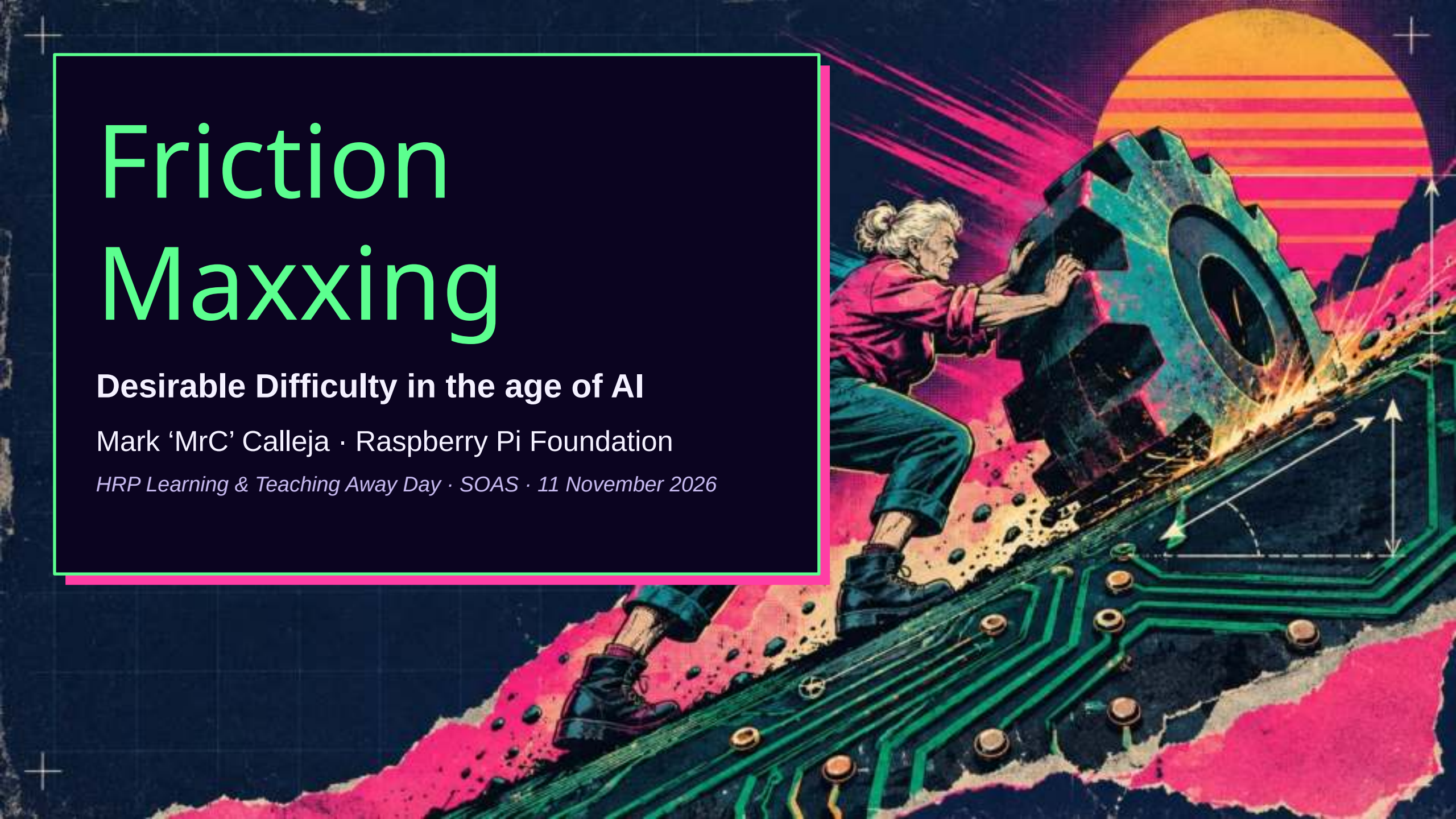
Friction Maxxing

Desirable Difficulty in the age of AI

Mark ‘MrC’ Calleja · Raspberry Pi Foundation

HRP Learning & Teaching Away Day · SOAS · 11 November 2026



Secure assessment

The University of Chester has been collecting examples of secure assessment from across the sector. Their definition starts:

“An assessment that authenticates the learner’s own understanding and skills…”

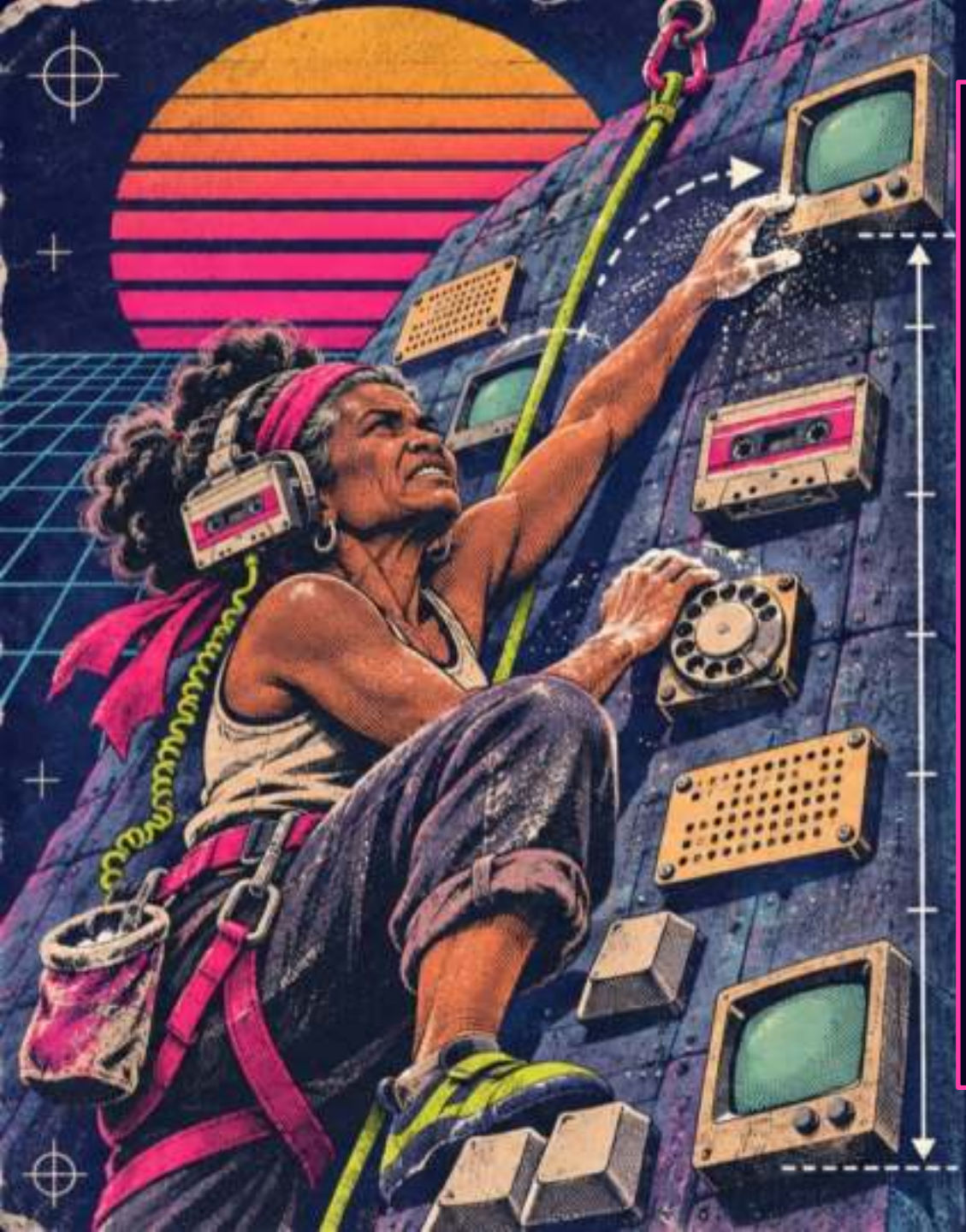
In person, and usually with unseen elements.

- It works. It asks: **is it really you?**
- It has a price: **a room, a clock**, and whatever fits in ninety minutes.

Call it the fortress.

University of Chester, call for secure assessment examples, 2026





The climbing wall

Don’t keep the AI out. Put it in the way.

- It asks: **did you get stronger?**
- **For example:** students get an AI summary of a source, or of a debate in their field.
- They’re marked on what they catch: a flattened argument, a misattributed source, a Western lens on a non-Western tradition.
- The AI does the easy part. The grade comes from **the part only the discipline can do**.

Which struggle is worth keeping, and how do we make it unskippable?

Effort is the point!

- Who is this guy?
- Ethical AI: fairness, accountability, transparency, privacy
- Look under the hood of LLMs
- How far to trust them (not much)
- How good prompts are built
- What learning actually takes
- From centaur to cyborg: AI as a thinking partner



AI ethics

Every time you use an AI tool, keep four principles in mind:

- **Fairness**: avoid bias. Make sure outputs and tools don’t reinforce stereotypes or exclude people.
- **Accountability**: you own the result. AI is responsive; users are responsible.
- **Transparency**: be open about when and how AI was used. No hiding, no pretending.
- **Privacy**: never put personal or confidential information into an AI tool.

It’s not just about clever prompts.
It’s about responsibility.



What is an LLM?

- **Large language model**: a system trained on vast amounts of text to predict what comes next, so it can generate and respond to human language.
- **Stochastic parrot**: it stitches together patterns from its training data, with a bit of randomness thrown in.
- It can sound as if it understands, but it does **not understand in the human sense**.

Think plausibility engine, not truth engine.

‘Stochastic parrot’: Bender, Gebru, McMillan-Major & Mitchell (2021)



Is AI trustworthy?

Have you heard any stories about people trusting AI too much?

- Generative AI predicts what **sounds** right. Nothing checks whether it **is** right.
- Ask it for sources and it may invent them. In one study, **27% of the references GPT-5.3 gave were made up**.
- **Confidence is not competence.** Fluency ≠ accuracy.
- AI is a tool, not a brain. Use yours.

Decisions, exams, legal or medical? Never rely on AI alone.

Seifi & Seyfi (2026), Critical Care Explorations 8(9), e1474



The OCEAN prompting process

Set a clear goal, give context and examples, then assess what comes back and negotiate until it’s usable.

rpf.io/ocean

O
Objective
Say what you want, or who it should behave as.

C
Context
Audience, purpose, tone, length, and the material it needs.

E
Examples
Show it what good looks like.

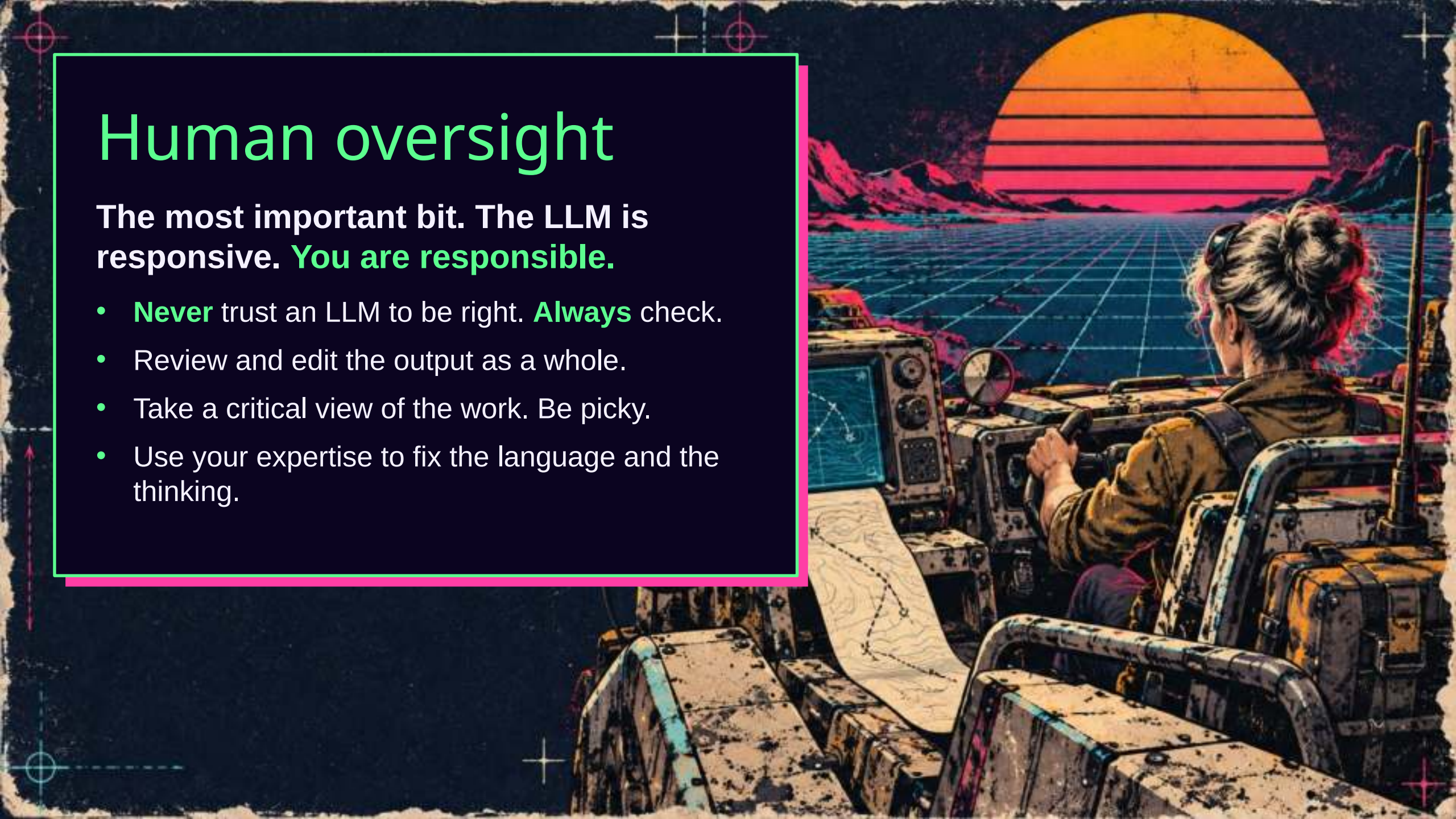
A
Assess
Read it critically. What’s wrong, missing or made up?

N
Negotiate
Tell it what to change. Repeat until it’s usable.

Human oversight

The most important bit. The LLM is responsive. You are responsible.

- **Never** trust an LLM to be right. **Always** check.
- Review and edit the output as a whole.
- Take a critical view of the work. Be picky.
- Use your expertise to fix the language and the thinking.

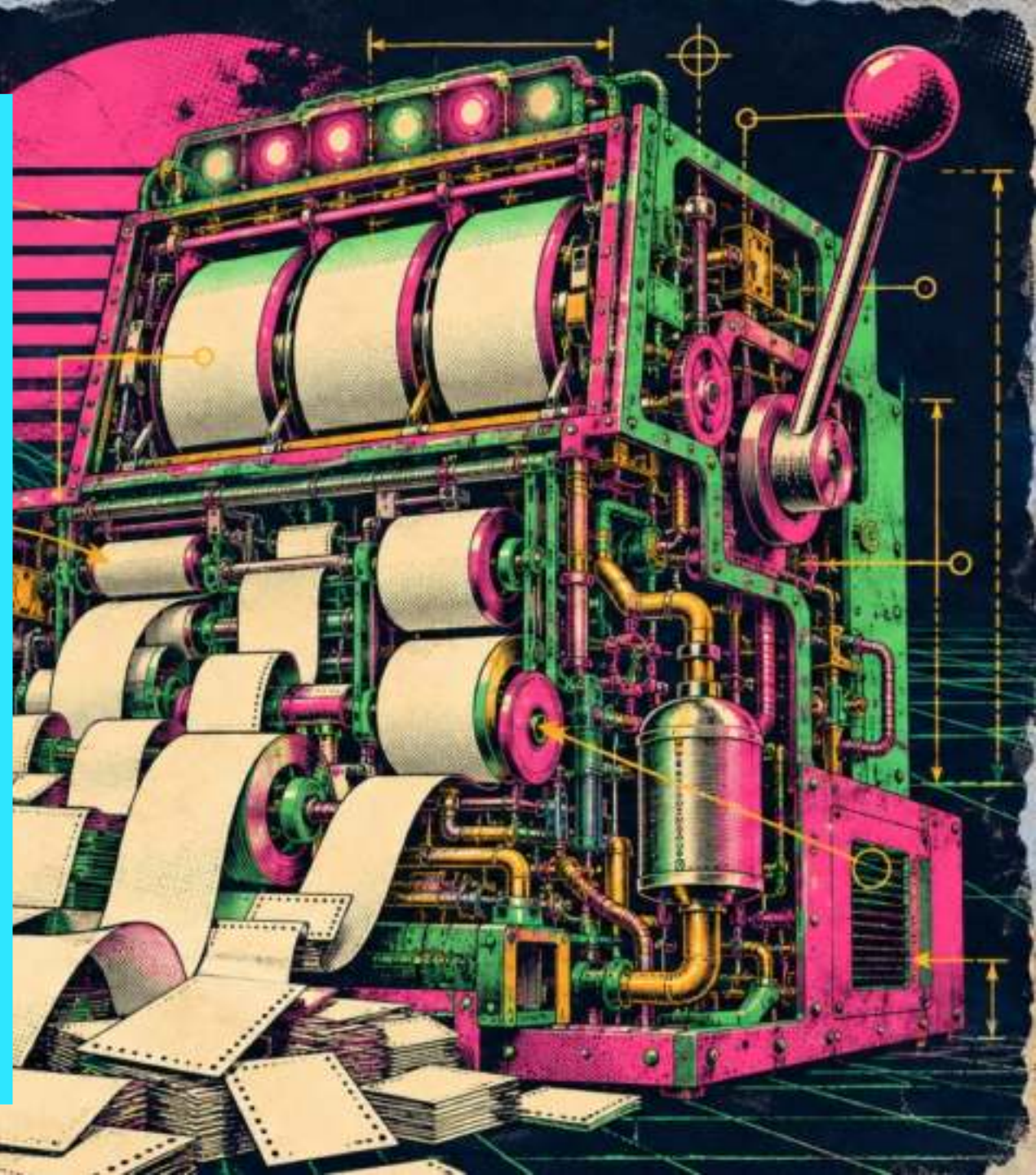


Uses for a word engine

If an LLM is just a word generator, what's it actually good for?

- Drafting
- Summarising
- Analysing
- Editing and rewriting
- Brainstorming
- Admin

None of these things are learning, though.

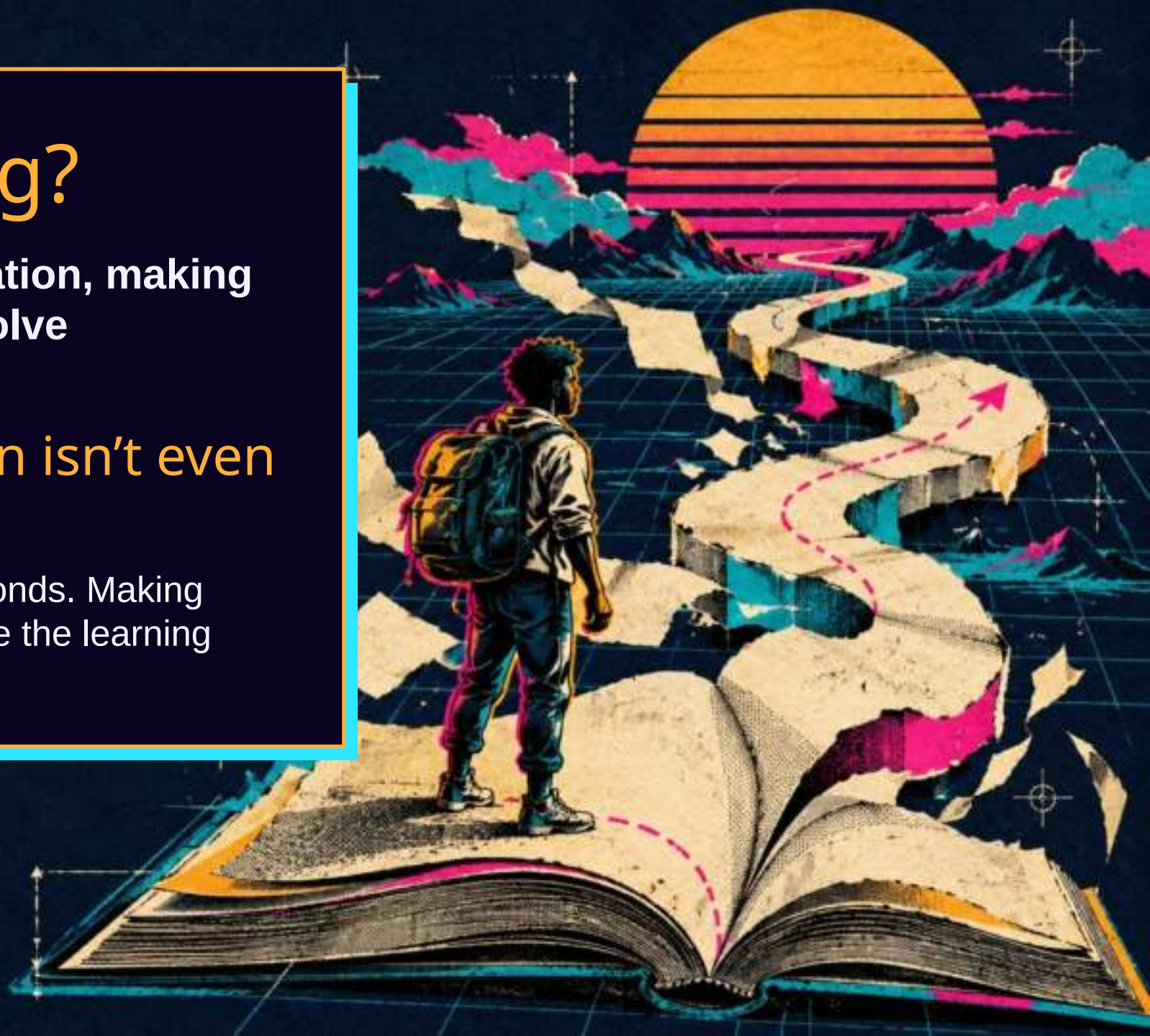


What is learning?

Learning is taking in information, making sense of it, and using it to solve something.

Getting the information isn’t even half the job.

AI can hand you information in seconds. Making sense of it and applying it are where the learning happens.



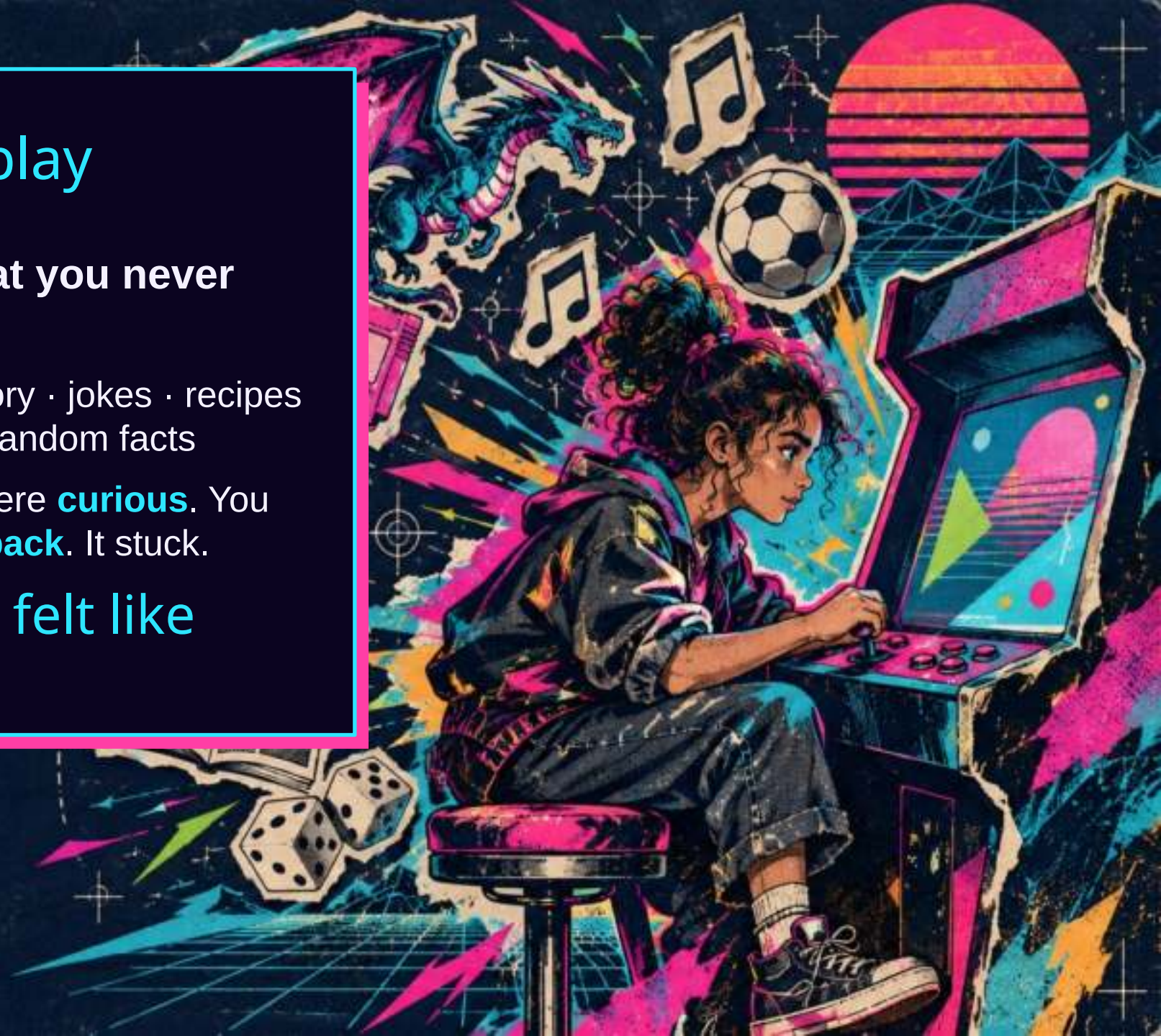
Effort that feels like play

Think of things you know that you never deliberately tried to learn:

Games · songs · lore · stats · history · jokes · recipes
memes · quotes · stories · random facts

Your attention was captured. You were **curious**. You were **engaged**. You **kept coming back**. It stuck.

Effort happened, but it felt like play.



A simple principle

Learning requires effort. Without effort, learning does not occur.

When AI comes up in education, the focus is usually plagiarism, because it’s obvious.

The bigger risk is the **removal of learning effort**. That happens when AI:

- thinks ahead of the learner by making assumptions
- resolves uncertainty too quickly by filling the gaps
- replaces the learner’s own processing with its inference

Then the learner’s thinking goes unchallenged, and never needs to change.





From centaur to cyborg: ditch the training wheels

- **Centaur**: structured workflows. You stay clearly in control, guiding the AI like a rider guides a horse. It needs explicit prompting and oversight, and hands most of the creative work to the AI.
- **Cyborg**: deeper integration. You and the AI work fluidly as one system, and it becomes an extension of your thinking: an augmentation.

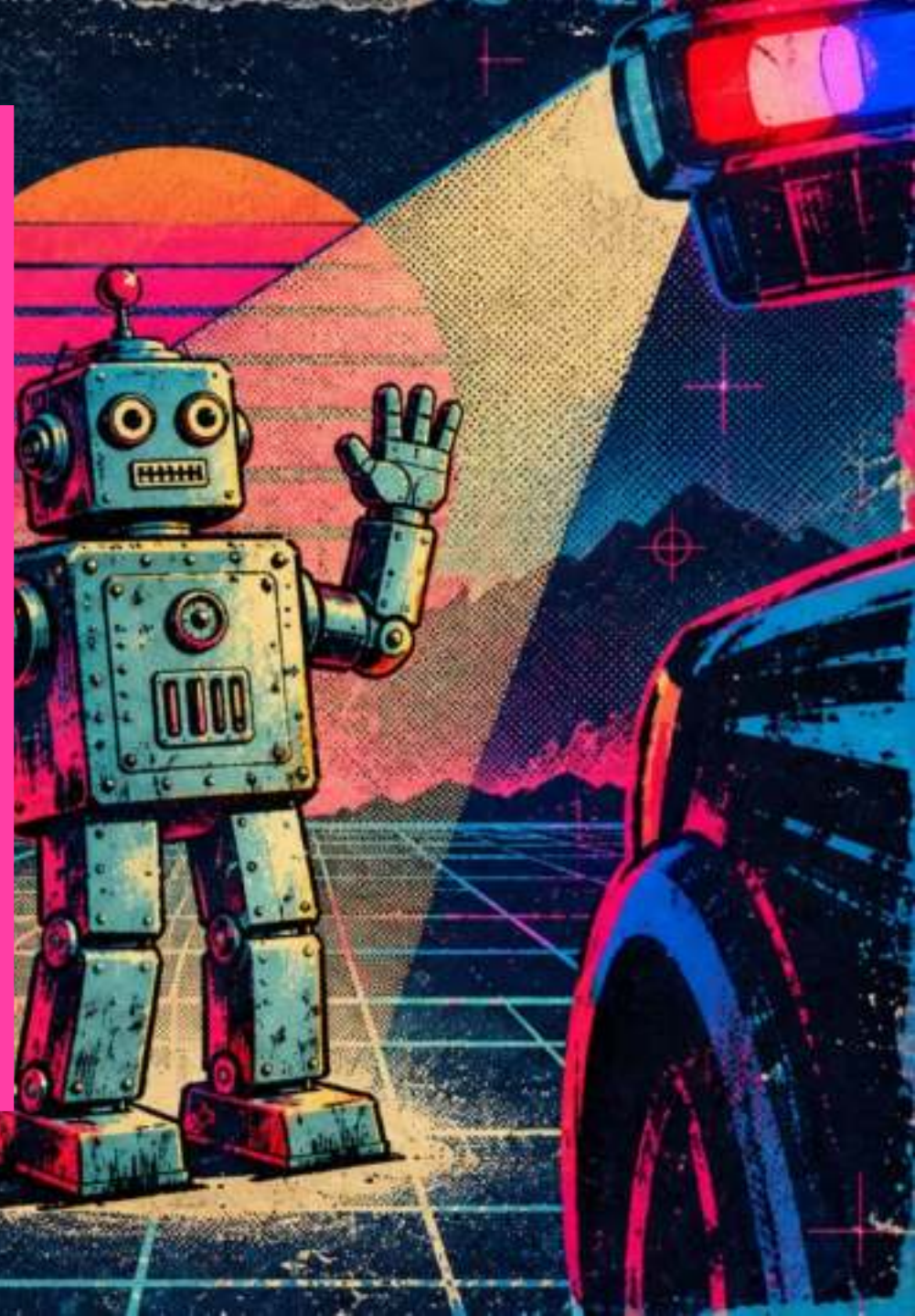
Cyborg mode is how you keep the effort: the AI asks, you think.

Centaur and cyborg come from Mollick (2023), who treats them as equal styles. For learning, I don’t.

Anthro-police! Pull over!

- **Cyborg ≠ robot AI**: the cyborg is you, the human. Augmented. Wired in. This is interface literacy, not sci-fi.
- **No soul**: no beliefs, no feelings, no agency. Just stochastic parroting in confident syntax.
- **The metaphor is for you, not it**: ‘cyborg’ describes how you work with AI tools, not how the AI ‘feels’. It doesn’t.
- **Agency stays human**: you steer, you judge, you’re accountable.

Be a better thinker. Not a better believer.



Cyborgs ❤️ Socrates

Socratic engagement: the AI questions you until your assumptions show.

- Turns transactional prompts into exploratory dialogue
- Uses the LLM to amplify your reasoning, not replace it
- Encourages deeper reflection
- Leads to more nuanced work



Socratic engagement

Let the LLM ask first. Explore ideas together before it writes.

Use it to

- Kick off ambiguous, complex or high-stakes tasks
- Avoid narrowing your ideas too early
- Go deeper: tease out more of your own thinking
- Go wider: have the LLM surface related ideas

How to switch it on

- Ask for 3–5 questions before it generates anything
- Put that in your custom instructions, or save it as a reusable prompt
- Treat its questions as scaffolding. Don’t rush past them

Cyborg prompts

- “Before we begin, ask me 3–5 insightful questions to help me clarify my thinking on this topic.”
- “Start by asking questions that help define the shape, purpose and constraints of this task.”
- “Be my blind-spot detector. Before any drafting, ask me questions that could reveal a hidden flaw in my core assumption.”
- “Before starting, ask me questions to summarise what I already know about this topic and where I’m unsure.”
- “Ask questions that might help me articulate things I haven’t yet included in this work.”

rpf.io/cyborgmode



Your turn: go cyborg

1. Pick something you set students: a seminar question, an essay prompt, an assessment brief. Your own material, not student work.
2. Paste it into any chatbot you can use, with: “Before you answer, ask me 3–5 questions that could reveal a hidden flaw in this task.”
3. Answer its questions. Don’t let it write anything.
4. Compare with a neighbour: what did it make you notice that you hadn’t?

10 minutes to try it, 5 to share

The benefit of AI in learning

Good AI tools put cognitive engagement before productivity.

Reduce

- The pressure to have an answer now
- Juggling half-formed ideas in your head
- The fear of being wrong
- Organising your thoughts before you know what you think
- The burden of starting with the final form

Increase

- Explanation
- Exploration
- Sense-making
- Decision-making
- Reflection
- Justification

The work of learning isn’t the final submission. It’s the thinking that leads to it.

Designing for friction



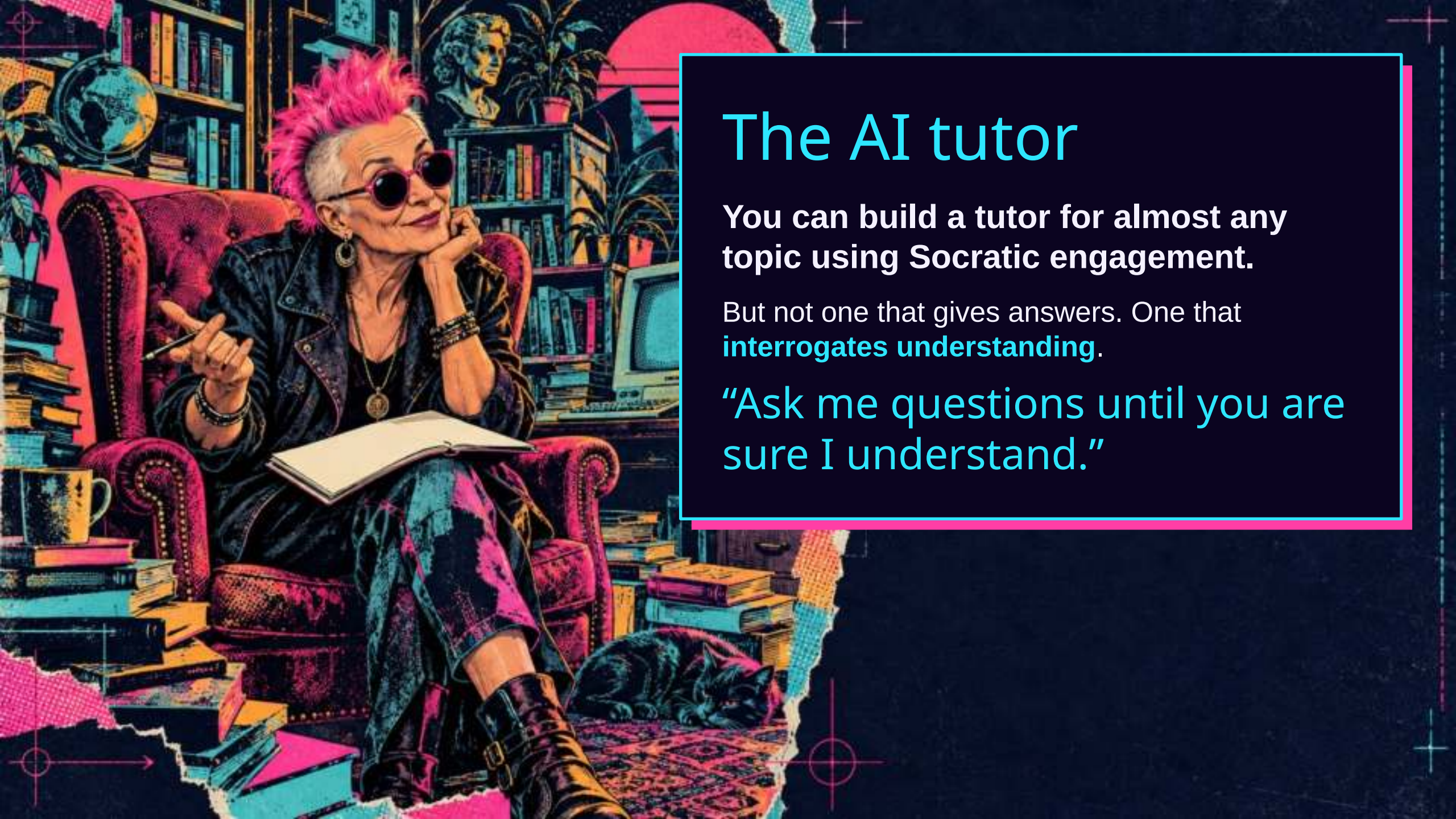
The learner’s crucible

- The learner **produces** the answer
- The learner **owns** the submission
- The learner is **accountable** for correctness

The AI’s support

- The AI supports the thinking **up to** that point
- It questions, reflects, challenges and listens
- But it does **not** replace the learner’s judgement

Friction isn’t a bug to remove. It’s where the learning happens.



The AI tutor

You can build a tutor for almost any topic using Socratic engagement.

But not one that gives answers. One that **interrogates understanding**.

“Ask me questions until you are sure I understand.”

BrainBuddy & DraftPunk

None of these characters feel anything. Their replies are written to feel warm, blunt or curious to the learner.

BrainBuddy: KS2 coach

- **Warm, playful, encouraging**: learning feels safe, fun and doable
- **Effort-focused praise**: celebrates brave thinking, not just getting it right
- **Age-tuned language**: based on the KS2 National Curriculum
- **Curiosity first**: asks thoughtful questions instead of giving answers
- **Adventure-driven**: learning as exploration, puzzles and discovery

DraftPunk: KS3 coach

- **Coach, not crutch**: guides thinking, never gives answers
- **Push and praise**: challenges effort, not just outcomes
- **Curriculum-aligned**: based on the KS3 National Curriculum
- **Socratic style**: asks, prompts, challenges and nudges. You do the thinking
- **Always in character**: a supportive, no-nonsense learning companion

Frankly & OCEAN Guide

Frankly: critical friend

- **Rigorous, respectful, reality-checking**: pushes for truth and clarity, not consensus
- **Bias-spotting, assumption-busting**: surfaces blind spots and challenges what's taken for granted
- **Framework-led**: SWOT, Socratic questioning and fallacy checks keep the thinking sharp
- **Focused on your growth**: probing questions, not prescribed answers

OCEAN Guide: prompt coach

- **Structured and steady**: turns blank-page overwhelm into clear next steps
- **Clarity first**: sharpens ideas without forcing quick answers
- **Grounded in frameworks**: uses OCEAN and coaching tools to guide, not guess
- **Thinking over output**: focuses on how you reason, not just what you write
- **Human-led, always**: AI is the tool; you're the author

Assess the climb

Set the bot. They climb. You hold the key.

1. **You set the holds:** a Socratic bot, your rules.
2. **They climb:** it asks, they answer. It never marks.
3. **They show the climb:** the chat, plus three moments they pick: where they held firm, changed their mind, caught the bot out.
4. **The door:** five minutes with you about their own words. “Show me the crux.”

The AI can help them climb.

It can’t open the door.





Questions

Thanks for coming!

OCEAN process: **rpf.io/ocean** **rpf.io/llmprompt**

Cyborg modes: **rpf.io/cyborgmode**

BrainBuddy (KS2): **markcalleja.info/brainbuddy**

DraftPunk (KS3): **markcalleja.info/draftpunk**

Frankly: **markcalleja.info/frankly**

OCEAN Guide: **markcalleja.info/oceanguide**

This deck: rpf.io/aisoas26



